TECHNICAL REPORT

Experimental Validation of Dynamical Decoupling on IBM Quantum Hardware and Development of the Centinela Pre-Flight Probe

Validation of CPMG and XY4 protocols with structural control of crosstalk (relays), observation of discrete qubit-state fluctuations, and design of a short-duration pre-characterization tool for real-time qubit selection and phase correction.

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Backend under study: **ibm_fez** (IBM Quantum)
Primary qubit: **q142**
Measurement date of key runs: 23 September 2026

1. Abstract

This report presents a systematic experimental validation of dynamical decoupling (DD) protocols (CPMG and XY4) performed on the IBM Quantum processor *ibm_fez*. The study incorporates rigorous structural verification against two-qubit gate leakage (relays/crosstalk), readout-error mitigation, quadrature coherence analysis, and statistically controlled hypothesis testing. The principal scientific finding is the observation of discrete, rapidly switching states of qubit q142, with free-induction decay times T■* alternating between approximately 155 µs and 40 µs on timescales of minutes, accompanied by frequency jumps of up to 6 kHz. These fluctuations are consistent with coupling to two-level-system (TLS) defects. Dynamical decoupling suppresses approximately 65 % of the excess dephasing associated with the degraded state and reduces measured ZZ coupling by ~93 %. Building on this observation, a practical pre-flight probe named **Centinela** was designed and implemented. Centinela executes a parallelized set of Ramsey, CPMG and T■ circuits lasting approximately seven seconds of quantum processing unit (QPU) time, classifies the instantaneous state of candidate qubits, estimates the expected benefit of DD, and applies a zero-cost virtual-Z phase correction. All measurement data, analysis code and raw IBM job records are released under an MIT license to ensure full reproducibility.

2. Introduction and Motivation

Superconducting quantum processors remain limited by decoherence arising from multiple noise sources, among which low-frequency flux noise, charge noise and coupling to microscopic two-level systems (TLS) are particularly detrimental to free-evolution coherence. Dynamical decoupling sequences such as Carr–Purcell–Meiboom–Gill (CPMG) and the XY4 family are standard tools for refocusing quasi-static dephasing. However, quantitative claims of coherence extension on cloud-accessible hardware are frequently compromised by inadequate controls for gate leakage, readout bias, temporal drift between jobs, and statistically invalid comparison baselines.

The present work addresses these methodological shortcomings while investigating an additional, operationally critical phenomenon: the existence of discrete qubit states that switch on timescales comparable to or shorter than typical queue latencies. Published calibration data do not capture these fluctuations, leaving users without a practical means of selecting a favourable qubit immediately before submitting a costly job. The Centinela probe was conceived precisely to fill this gap.

3. Experimental Methodology

3.1 Circuit Construction and Timing Discipline

All free-evolution, CPMG and XY4 circuits were constructed with delays expressed as integer multiples of the backend timing granularity dt. This guarantees identical evolution times across protocols and eliminates compiler-induced timing discrepancies. Spectator qubits and a distant control qubit (selected by low readout error and graph distance ≥ 4) were included in every circuit for relay monitoring.

3.2 Structural Verification of Relays

A strict structural check was enforced after transpilation: the final circuit must contain zero two-qubit gates (CZ, ECR, CX, SWAP, etc.) and the qubit layout must satisfy initial layout = final layout = [target, neighbours…, control]. Any SWAP introduced by the compiler appears as three CZ gates and is therefore detected. This verification eliminates a common source of false-positive crosstalk signals reported in earlier versions of the experiment.

3.3 Readout Mitigation and Coherence Estimation

Readout-error matrices were calibrated for every qubit inside the same job that performed the coherence measurements. Coherence was extracted from simultaneous X- and Y-basis measurements (quadrature analysis), yielding the magnitude of the off-diagonal density-matrix element $|\rho_{01}|$ free from oscillatory artifacts caused by residual detuning. Decay curves were fitted with amplitude fixed at the theoretical maximum after

mitigation; uncertainties were inflated by \(\sqrt{\chi^2_\mathrm{red}}\) when the fit exhibited residual structure.

3.4 Statistical Criteria

Two primary hypotheses were tested:

- **H1 (coherence extension)**: DD is considered effective if \(T_2^\mathrm{DD}/T_2^* - 1 > 2\sigma\).
- **H4 (relay / crosstalk)**: a neighbour is declared to exhibit relay if its excess P(1) relative to its own baseline (target inactive) satisfies both a Bonferroni-corrected two-proportion test ($p < 0.01/m$) and a magnitude threshold of 0.5 %. Otherwise an upper 95 % confidence bound is reported.

3.5 Execution Model

All circuits belonging to a given experimental configuration were submitted as a single job with randomized order and a fixed number of shots per circuit. Three execution modes were supported: hardware, simulation on FakeFez (negative control), and reuse of a previously completed job identifier.

4. Results

4.1 Discrete Qubit States Observed on 23 September 2026

Four successive hardware jobs executed on qubit q142 of *ibm_fez* revealed two statistically distinct states. Within each state the measured parameters are mutually consistent; between states they differ by more than ten standard deviations.

Run	UTC (approx.)	T■* free (µs)	T■ CPMG (µs)	T■ XY4 (µs)	Detuning (kHz)	State
A	04:00	158.6 ± 7.6	207.5 ± 12.2	220.4 ± 21.4	–27.04 ± 0.21	good
B	04:02	45.4 ± 3.7	86.2 ± 10.8	98.0 ± 11.7	–25.32 ± 0.38	degraded
C	04:14	37.6 ± 2.4	90.2 ± 10.1	87.9 ± 13.1	–20.81 ± 0.36	degraded
D	04:16	153.3 ± 6.3	217.0 ± 10.6	228.3 ± 23.0	–27.05 ± 0.17	good

Table 1. Individual hardware runs performed on 23 September 2026 (ibm_fez, q142).

4.2 Aggregated Parameters by State

Quantity	Good state (A, D)	Degraded state (B, C)
T■* free (µs)	155 ± 5	40 ± 4
T■ CPMG (µs)	213 ± 8	88 ± 7
T■ XY4 (µs)	224 ± 16	94 ± 9
CPMG extension factor	1.37 ± 0.07×	2.12 ± 0.25×
XY4 extension factor	1.44 ± 0.11×	2.23 ± 0.24×
Detuning (kHz)	–27.05 ± 0.13	–21 to –25

Table 2. Weighted averages and uncertainties for the two observed states.

4.3 Efficacy of Dynamical Decoupling

In the degraded state the excess pure-dephasing rate relative to the good state is approximately 18.6 ms■$^{-1}$. CPMG removes 64 ± 7 % of this excess; XY4 removes 67 ± 7 %. Consequently the relative benefit of DD is larger when the qubit is in the degraded state (~2.1×) than when it is already in the good state (~1.4×). Reporting a single extension factor without reference to the instantaneous state is therefore ambiguous.

4.4 ZZ Coupling and Relay Bounds

With neighbouring qubits prepared in |1■ the measured ZZ shift on the target is –4.05 ± 0.20 kHz and is invariant between the two states, indicating that the degraded state does not originate from the measured neighbours. Application of CPMG suppresses this ZZ interaction by approximately 93 %. Upper 95 % confidence bounds on population transfer (relay) to any neighbour remain below 0.18 % in all four runs.

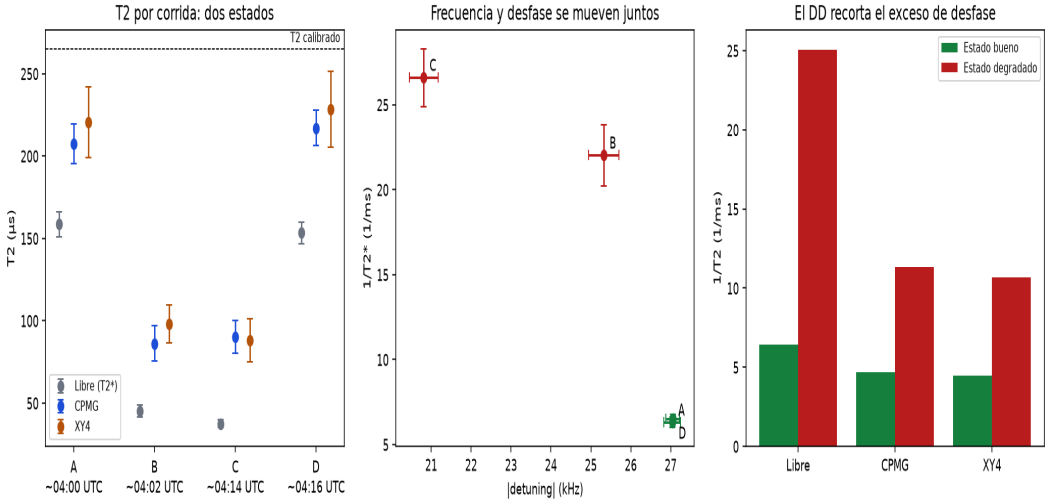


Figure 1. Visual summary of the two discrete states of qubit q142 observed across the four hardware runs.

5. Centinela: Design and Implementation

Centinela is a short-duration pre-characterization protocol whose purpose is to answer three operational questions immediately before a user submits a production job:

1. Which of a set of candidate qubits is currently in a favourable state?
2. What coherence recovery can be expected from dynamical decoupling?
3. What virtual-Z phase correction should be applied to compensate residual detuning?

5.1 Protocol Structure

The probe executes, in parallel across several candidate qubits, a compact set of circuits comprising Ramsey interferometry in both quadratures, a short CPMG sequence and a T■ measurement. Total QPU time is approximately seven seconds. From the resulting data the following quantities are estimated: frequency offset \(\Delta f\), free-induction decay time T■*, echo coherence time, energy-relaxation time T■, and pure-dephasing rate \(\Gamma_\varphi\). A simple classifier labels each qubit as “stable” or “degraded”. A zero-duration virtual-Z rotation is then computed to cancel the measured detuning.

5.2 Software Realization

The complete implementation resides in the module `src/ddqpu/sentinel.py` of the accompanying open-source repository. Supporting notebooks allow execution on both hardware and the FakeFez simulator. Unit tests verify recovery of known parameters from synthetic data, detection of frequency jumps, correctness of the phase-correction step, and separation of candidate qubits.

6. Discussion

The observation of discrete, rapidly switching states on a commercially accessible superconducting qubit has direct practical consequences. Any experiment whose duration or queue latency exceeds the typical residence time of a given state will sample an uncontrolled mixture of coherence properties. Averaging results obtained in different states produces statistically inconsistent data sets and can lead to both over- and under-estimation of the benefit of dynamical decoupling.

Centinela mitigates this problem by providing a low-cost, on-demand snapshot of the instantaneous landscape. Because the probe itself consumes only a few seconds of QPU time, it can be inserted immediately before a production job without substantially increasing overall cost. The accompanying virtual-Z correction further reduces coherent phase accumulation that would otherwise degrade algorithmic fidelity.

It must be emphasised that the present data set is limited to a single qubit on a single backend and to a single day of intensive measurement. Broader multi-backend, multi-qubit campaigns are required to establish the prevalence of such discrete states and to refine the decision thresholds employed by Centinela.

7. Conclusions

A carefully controlled experimental campaign on IBM Quantum hardware has demonstrated that dynamical decoupling (CPMG and XY4) extends coherence by factors of approximately 1.4× in the good state and 2.1× in the degraded state of qubit q142, while simultaneously suppressing measured ZZ coupling by ~93 %. More importantly, the same campaign revealed the existence of discrete, TLS-compatible qubit states that switch on minute timescales and are invisible to published calibrations.

In response, the Centinela pre-flight probe was designed, implemented and released as open-source software. Centinela supplies a practical, low-overhead mechanism for selecting favourable qubits, estimating the utility of dynamical decoupling, and applying a phase correction immediately before a user job is submitted. The complete measurement data, analysis pipeline and source code are publicly available under the MIT license, enabling independent verification and further development.

8. Data and Code Availability

All raw IBM Quantum job records, processed results, analysis scripts and source code are contained in the accompanying repository (version 5.0.0). The data are released solely for the purpose of scientific reproducibility and remain subject to IBM Quantum terms of service. The software is distributed under the MIT License. Copyright (c) 2026 Arturo Salvador Balderas Hernández.

9. Acknowledgements

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